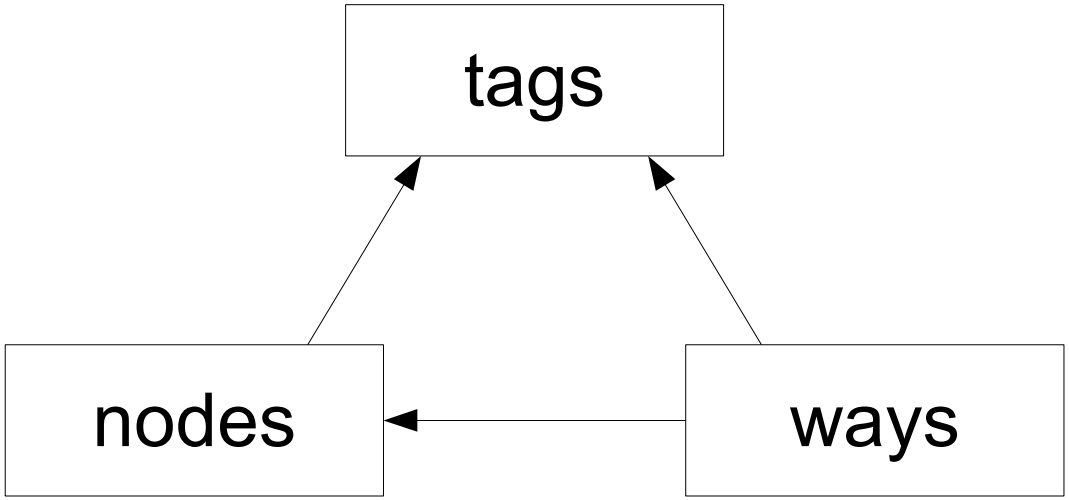
Relation Toolbox:

Editing Relations Should Be Easy

Ilya Zverev, SotM-EU 2011

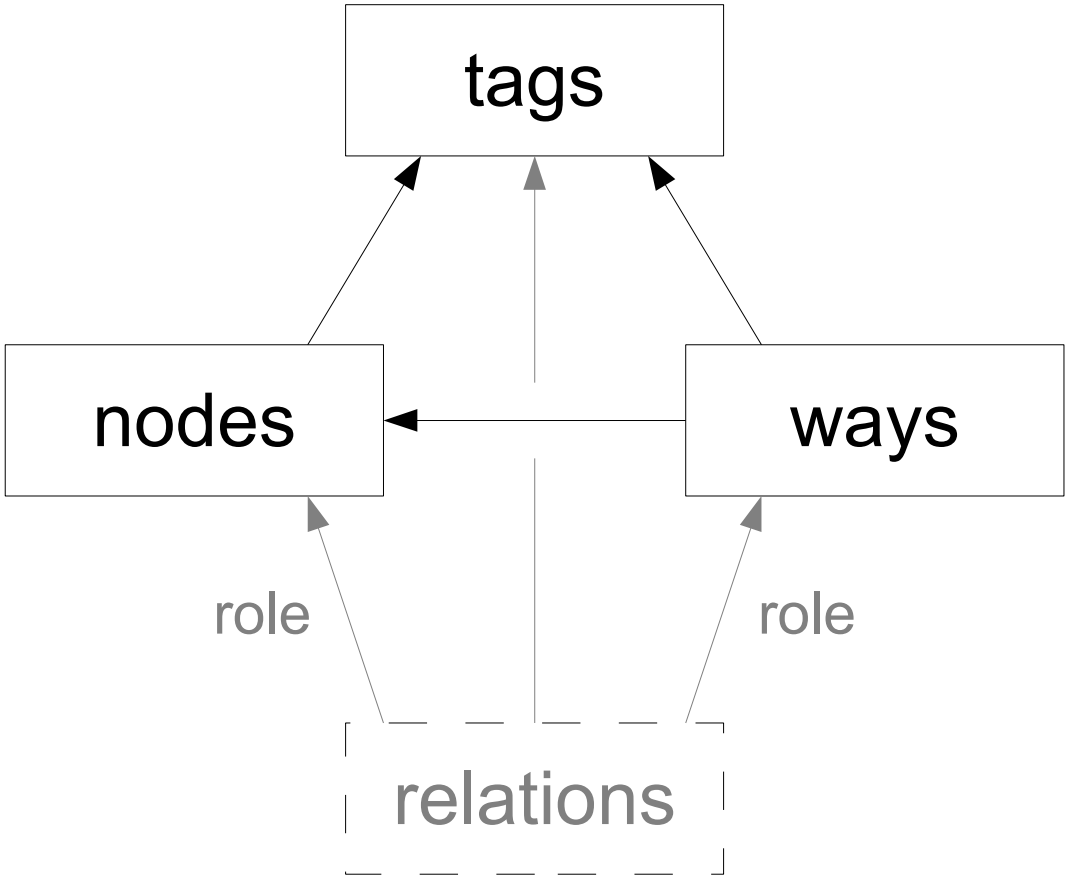
OpenStreetMap Data Model

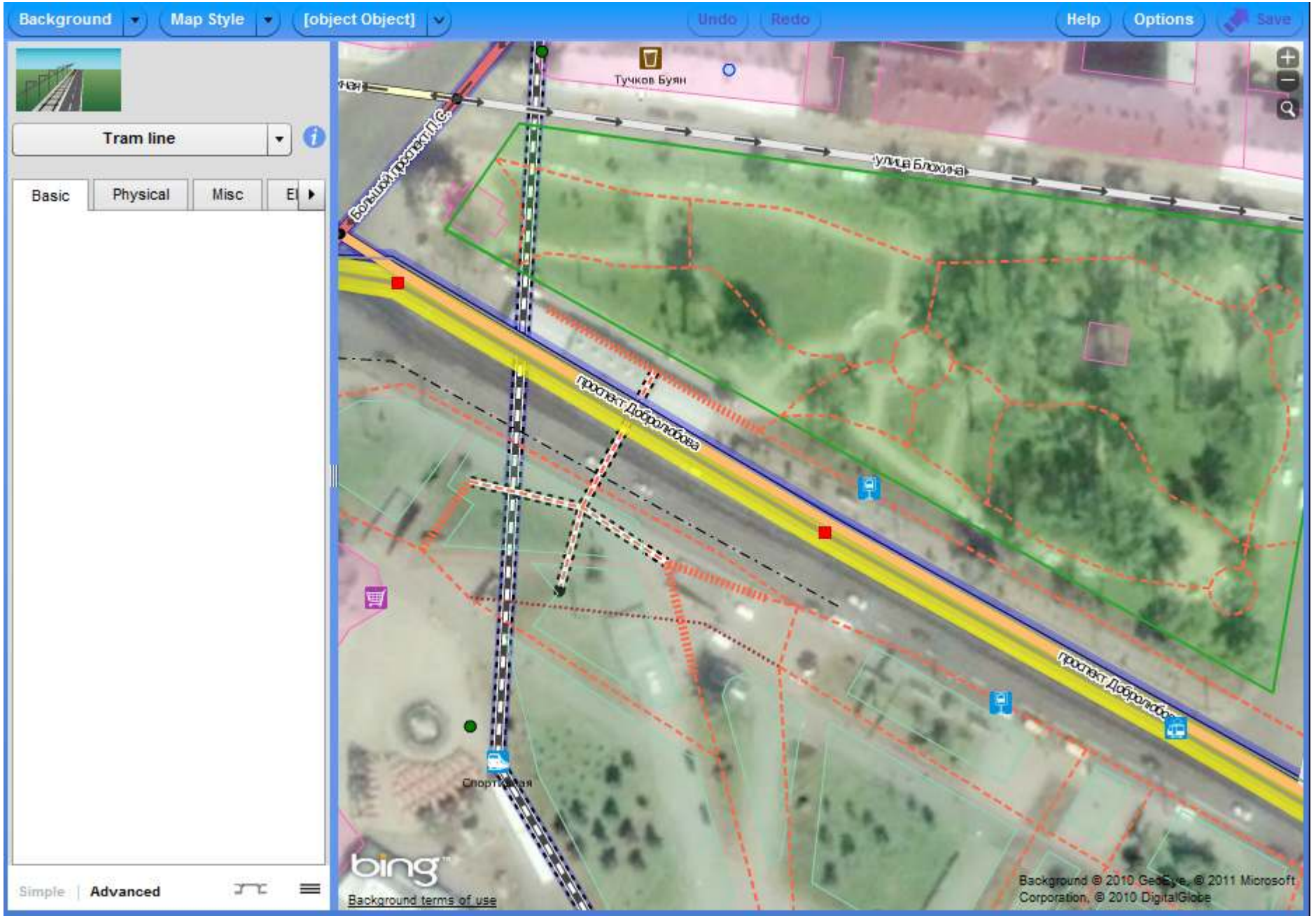
(as presented to other people)

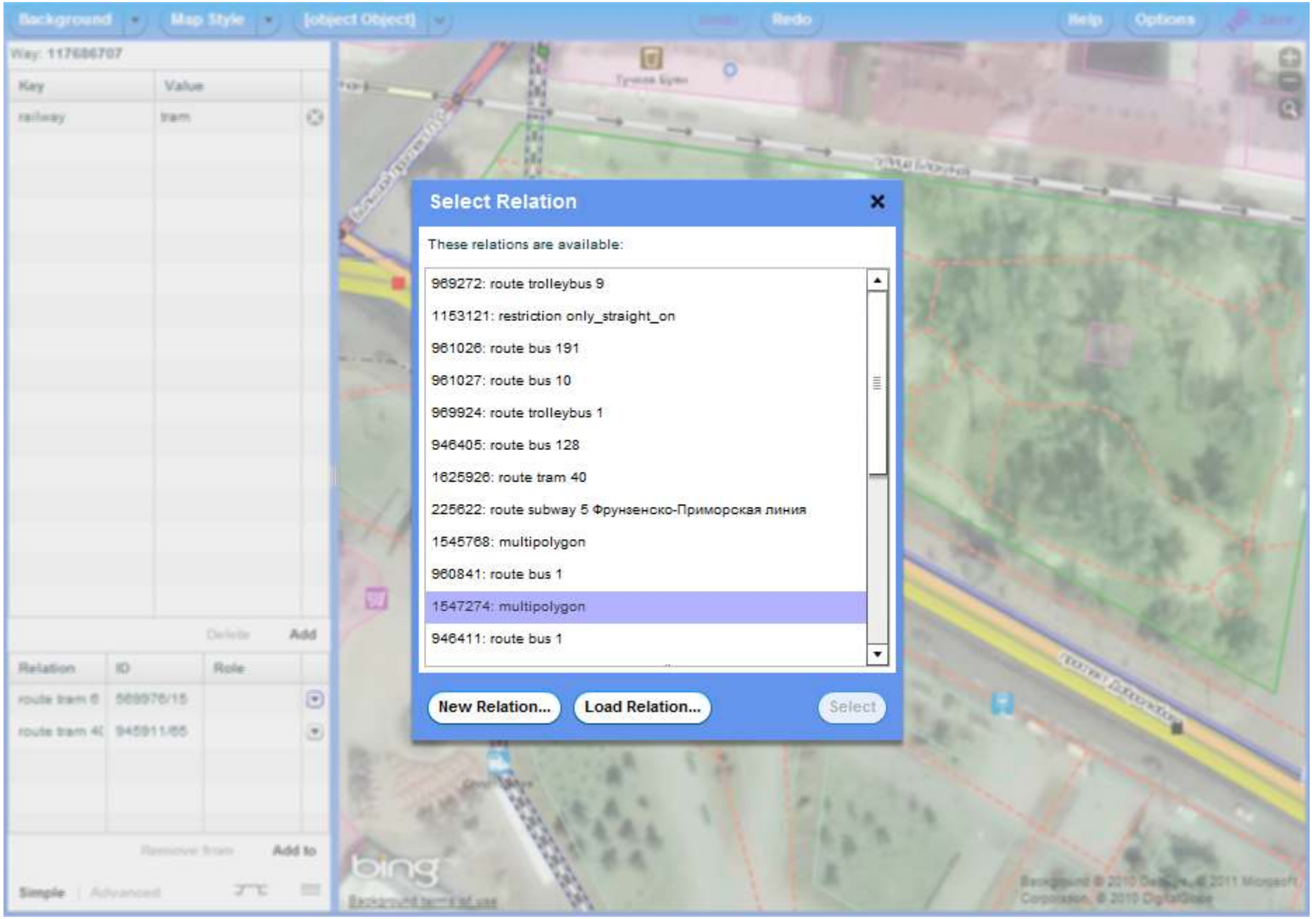


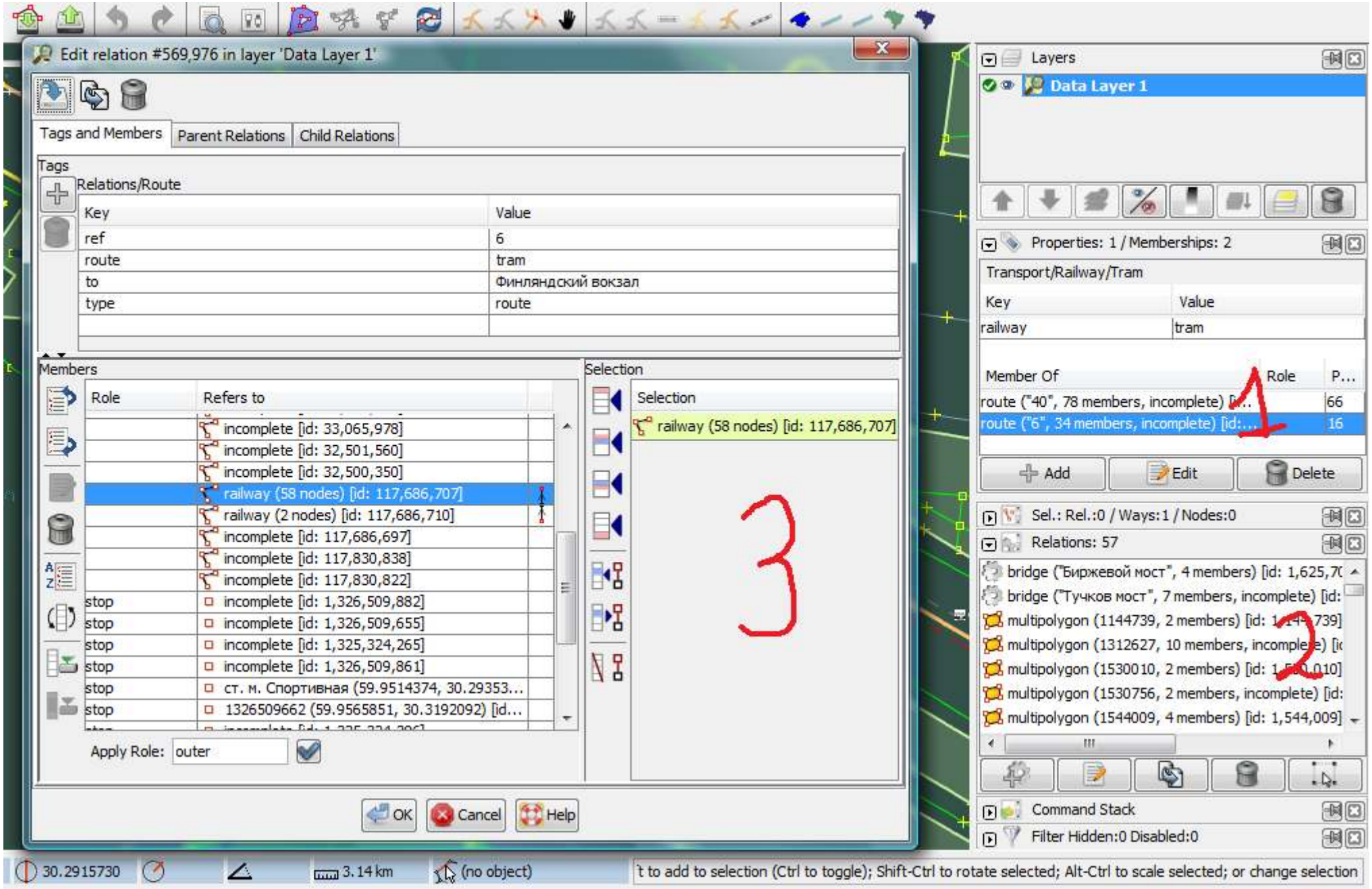
OpenStreetMap Data Model

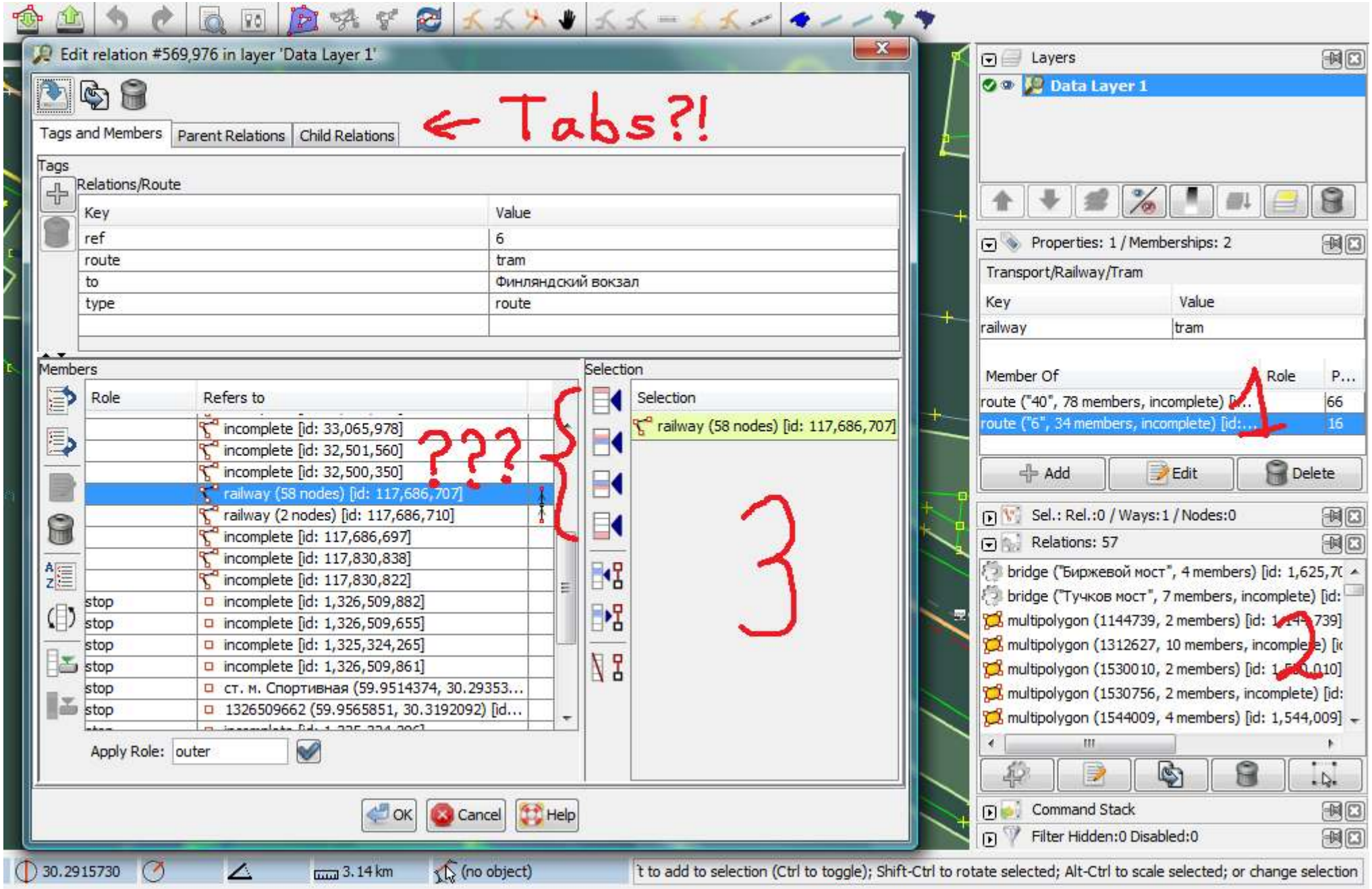
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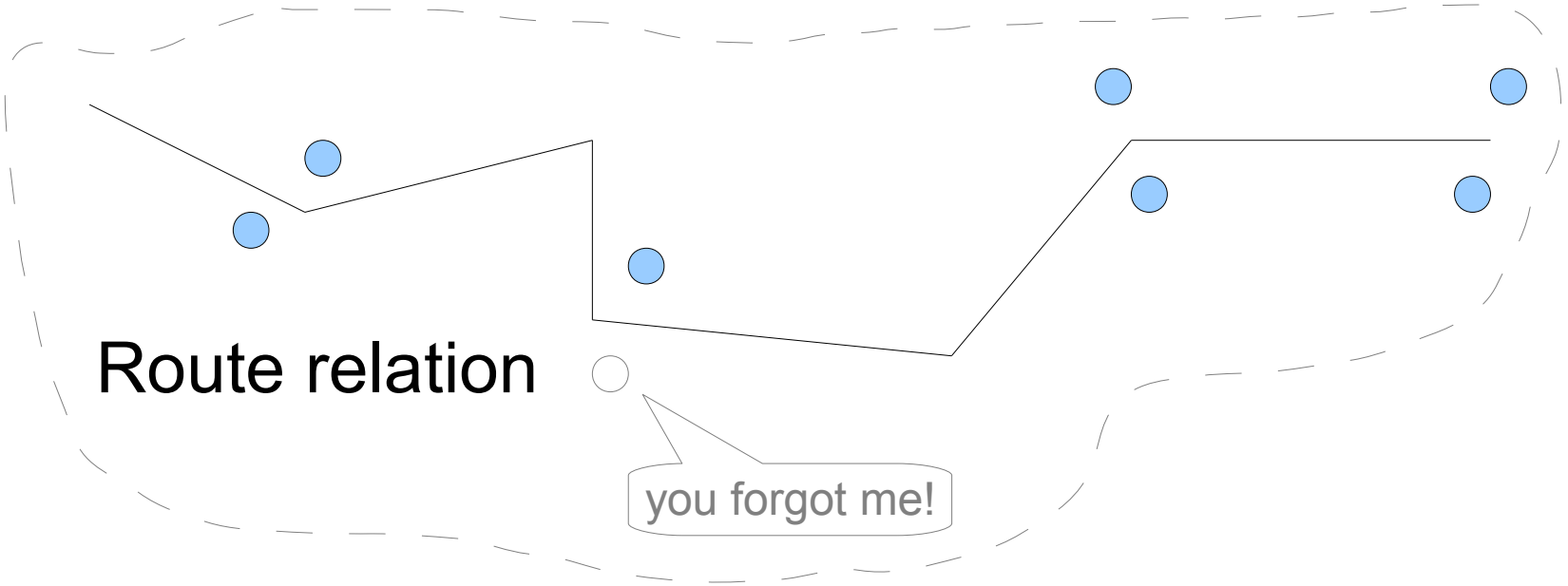


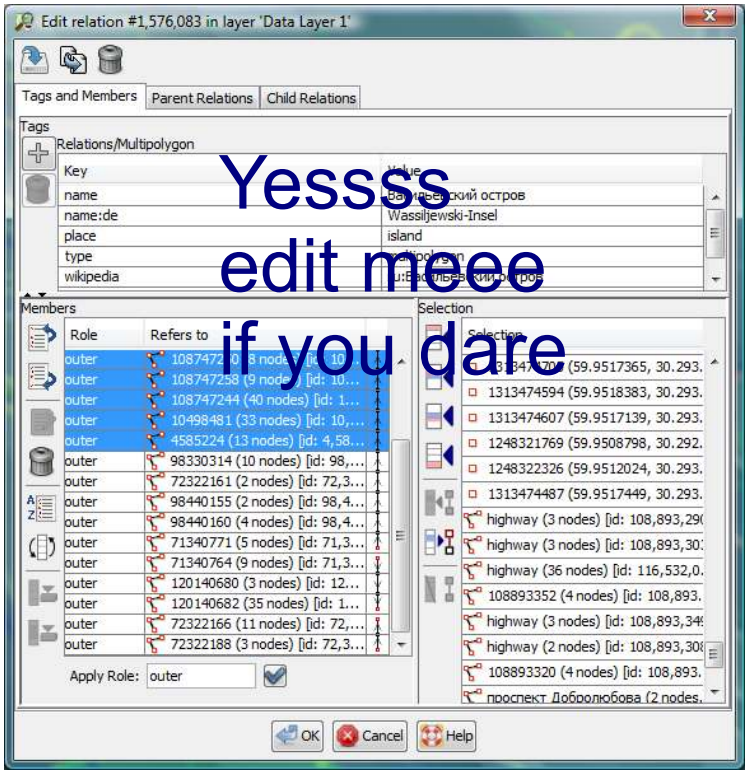

Background
Map Style
[object Object]
Undo
Redo
Help
Options
Save
Tram line
Basic
Physical
Misc
Simple
Advanced
Тучков Буян
улица Блохина
проспект Добролюбова
проспект Добролюбова
bing
Background terms of use
Background © 2010 GeoEye, © 2011 Microsoft Corporation, © 2010 DigitalGlobe


Background
Map Style
[object Object]
Undo
Redo
Help
Options
Save
Way: 117686707
Key
Value
railway
tram
Delete
Add
Relation
ID
Role
route tram 6
569976/15
route tram 4(
945911/65
Remove from
Add to
Simple
Advanced
Select Relation
These relations are available:
969272: route trolleybus 9
1153121: restriction only_straight_on
961026: route bus 191
961027: route bus 10
969924: route trolleybus 1
946405: route bus 128
1625926: route tram 40
225622: route subway 5 Фрунзенско-Приморская линия
1545768: multipolygon
960841: route bus 1
1547274: multipolygon
946411: route bus 1
New Relation...
Load Relation...
Select
bing


Edit relation #569,976 in layer 'Data Layer 1'
Tags and Members
Parent Relations
Child Relations
Tags
Relations/Route
Key
Value
ref
6
route
tram
to
Финляндский вокзал
type
route
Members
Role
Refers to
incomplete [id: 33,065,978]
incomplete [id: 32,501,560]
incomplete [id: 32,500,350]
railway (58 nodes) [id: 117,686,707]
railway (2 nodes) [id: 117,686,710]
incomplete [id: 117,686,697]
incomplete [id: 117,830,838]
incomplete [id: 117,830,822]
stop
incomplete [id: 1,326,509,882]
stop
incomplete [id: 1,326,509,655]
stop
incomplete [id: 1,325,324,265]
stop
incomplete [id: 1,326,509,861]
stop
ст. м. Спортивная (59.9514374, 30.29353...
stop
1326509662 (59.9565851, 30.3192092) [id...
Apply Role:
outer
Selection
Selection
railway (58 nodes) [id: 117,686,707]
3
OK
Cancel
Help
Layers
Data Layer 1
Properties: 1 / Memberships: 2
Transport/Railway/Tram
Key
Value
railway
tram
Member Of
Role
P...
route ("40", 78 members, incomplete) [i...
66
route ("6", 34 members, incomplete) [id:...
16
1
Add
Edit
Delete
Sel.: Rel.:0 / Ways:1 / Nodes:0
Relations: 57
bridge ("Биржевой мост", 4 members) [id: 1,625,7
bridge ("Тучков мост", 7 members, incomplete) [id:
multipolygon (1144739, 2 members) [id: 1,144,739]
multipolygon (1312627, 10 members, incomplete) [i
multipolygon (1530010, 2 members) [id: 1,530,010]
multipolygon (1530756, 2 members, incomplete) [id:
multipolygon (1544009, 4 members) [id: 1,544,009]
2
Command Stack
Filter Hidden:0 Disabled:0
30.2915730
3.14 km
(no object)
t to add to selection (Ctrl to toggle); Shift-Ctrl to rotate selected; Alt-Ctrl to scale selected; or change selection


Edit relation #569,976 in layer 'Data Layer 1'
← Tabs?!
Tags and Members
Parent Relations
Child Relations
Tags
Relations/Route
Key
Value
ref
6
route
tram
to
Финляндский вокзал
type
route
Members
Role
Refers to
incomplete [id: 33,065,978]
incomplete [id: 32,501,560]
incomplete [id: 32,500,350]
railway (58 nodes) [id: 117,686,707]
railway (2 nodes) [id: 117,686,710]
incomplete [id: 117,686,697]
incomplete [id: 117,830,838]
incomplete [id: 117,830,822]
stop
incomplete [id: 1,326,509,882]
stop
incomplete [id: 1,326,509,655]
stop
incomplete [id: 1,325,324,265]
stop
incomplete [id: 1,326,509,861]
stop
ст. м. Спортивная (59.9514374, 30.29353...
stop
1326509662 (59.9565851, 30.3192092) [id...
???
Apply Role:
outer
Selection
Selection
railway (58 nodes) [id: 117,686,707]
3
OK
Cancel
Help
Layers
Data Layer 1
Properties: 1 / Memberships: 2
Transport/Railway/Tram
Key
Value
railway
tram
Member Of
Role
P...
route ("40", 78 members, incomplete) [i...
66
route ("6", 34 members, incomplete) [id:...
16
1
Add
Edit
Delete
Sel.: Rel.:0 / Ways:1 / Nodes:0
Relations: 57
bridge ("Биржевой мост", 4 members) [id: 1,625,7(
bridge ("Тучков мост", 7 members, incomplete) [id:
multipolygon (1144739, 2 members) [id: 1,144,739]
multipolygon (1312627, 10 members, incomplete) [i
multipolygon (1530010, 2 members) [id: 1,530,010]
multipolygon (1530756, 2 members, incomplete) [id:
multipolygon (1544009, 4 members) [id: 1,544,009]
2
Command Stack
Filter Hidden:0 Disabled:0
30.2915730
3.14 km
(no object)
t to add to selection (Ctrl to toggle); Shift-Ctrl to rotate selected; Alt-Ctrl to scale selected; or change selection

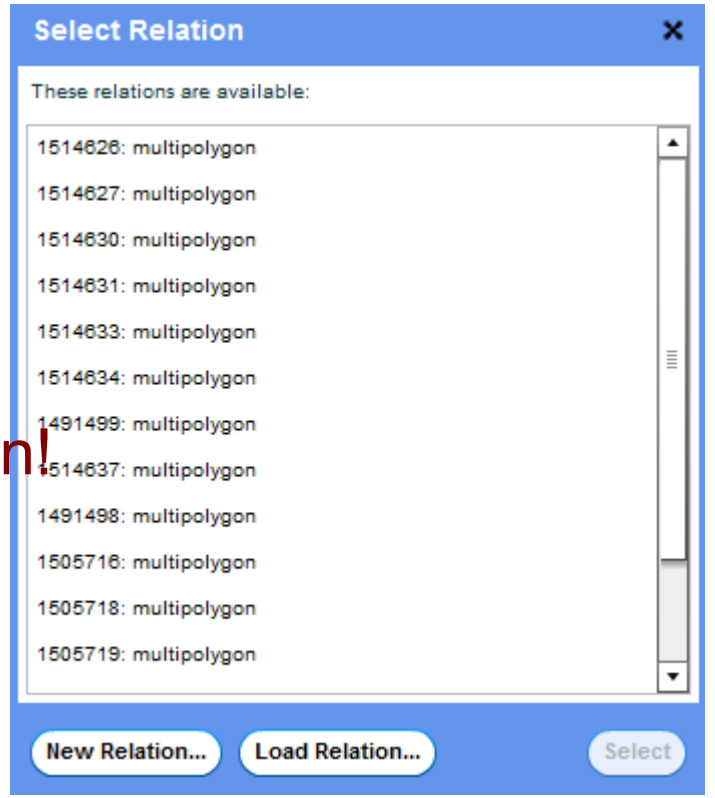


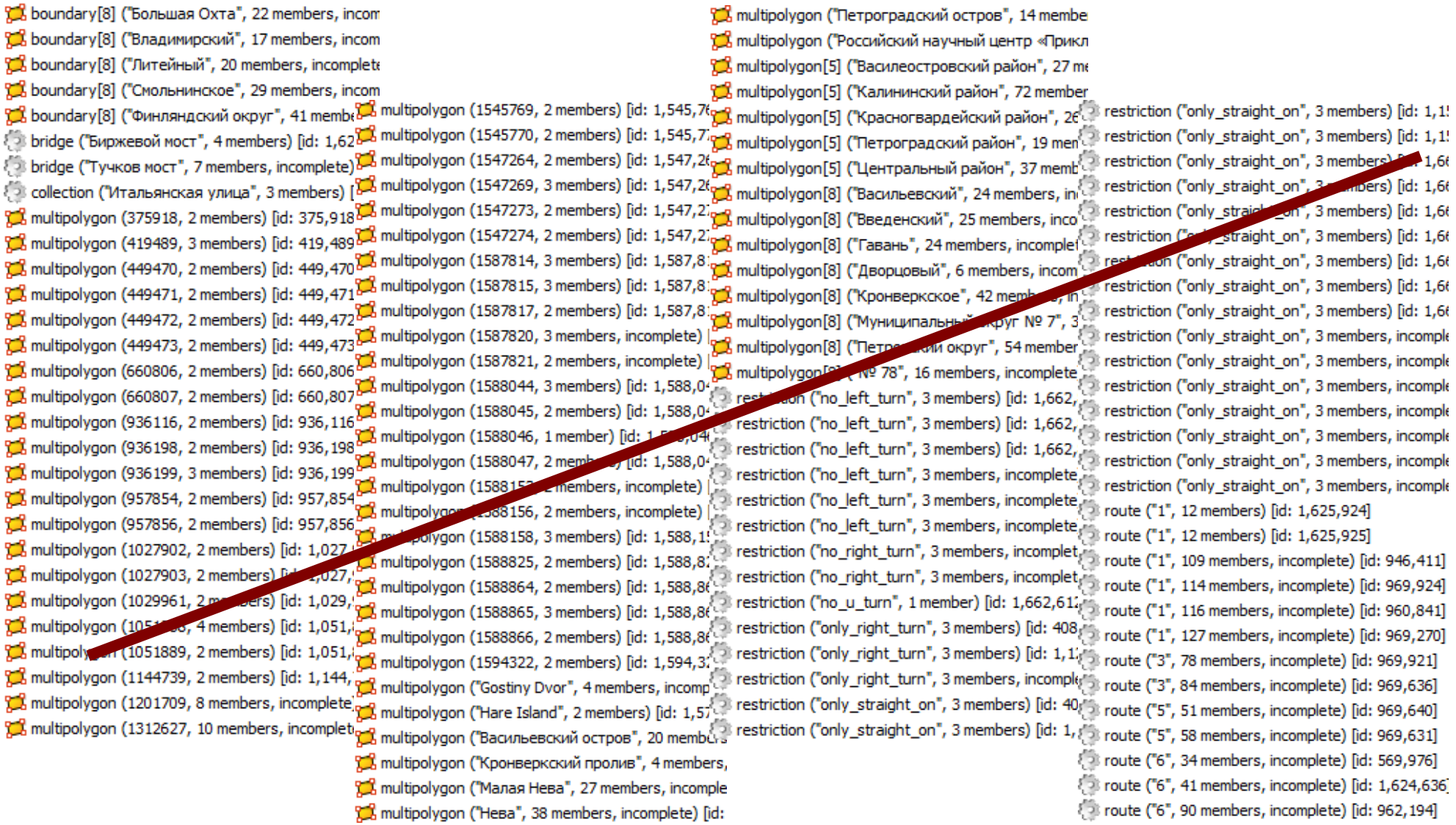


Yessss
edit meee
if you dare

You need
to add a hole
to a multipolygon!

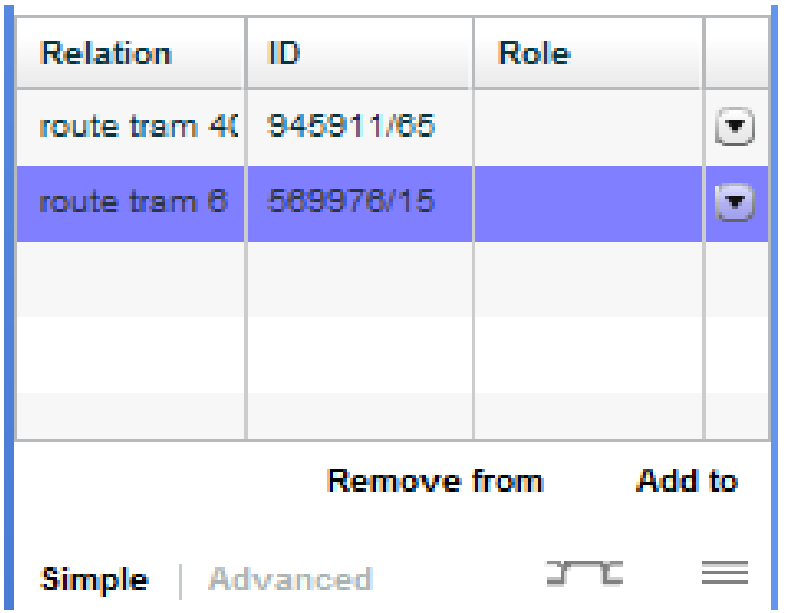
(any of them,
probably)

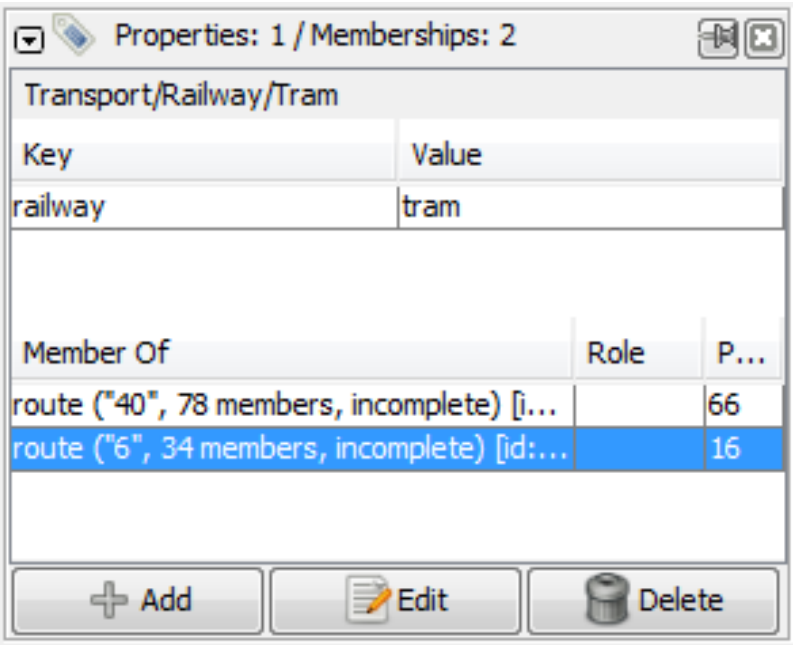



boundary[8] ("Большая Охта", 22 members, incom
boundary[8] ("Владимирский", 17 members, incom
boundary[8] ("Литейный", 20 members, incomplete
boundary[8] ("Смольнинское", 29 members, incom
boundary[8] ("Финляндский округ", 41 membe
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bridge ("Тучков мост", 7 members, incomplete)
collection ("Итальянская улица", 3 members) [
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multipolygon (419489, 3 members) [id: 419,489
multipolygon (449470, 2 members) [id: 449,470
multipolygon (449471, 2 members) [id: 449,471
multipolygon (449472, 2 members) [id: 449,472
multipolygon (449473, 2 members) [id: 449,473
multipolygon (660806, 2 members) [id: 660,806
multipolygon (660807, 2 members) [id: 660,807
multipolygon (936116, 2 members) [id: 936,116
multipolygon (936198, 2 members) [id: 936,198
multipolygon (936199, 3 members) [id: 936,199
multipolygon (957854, 2 members) [id: 957,854
multipolygon (957856, 2 members) [id: 957,856
multipolygon (1027902, 2 members) [id: 1,027,
multipolygon (1027903, 2 members)
multipolygon (1029961, 2 members) [id: 1,029,
multipolygon (1051 4 members) [id: 1,051,
multipolygon (1051889, 2 members) [id: 1,051,
multipolygon (1144739, 2 members) [id: 1,144,
multipolygon (1201709, 8 members, incomplete
multipolygon (1312627, 10 members, incomplet
multipolygon (1545769, 2 members) [id: 1,545,7
multipolygon (1545770, 2 members) [id: 1,545,7
multipolygon (1547264, 2 members) [id: 1,547,2
multipolygon (1547269, 3 members) [id: 1,547,2
multipolygon (1547273, 2 members) [id: 1,547,2
multipolygon (1547274, 2 members) [id: 1,547,2
multipolygon (1587814, 3 members) [id: 1,587,8
multipolygon (1587815, 3 members) [id: 1,587,8
multipolygon (1587817, 2 members) [id: 1,587,8
multipolygon (1587820, 3 members, incomplete)
multipolygon (1587821, 2 members, incomplete)
multipolygon (1588044, 3 members) [id: 1,588,0
multipolygon (1588045, 2 members) [id: 1,588,0
multipolygon (1588046, 1 member) [id:
multipolygon (1588047, 2 mem
multipolygon (1588156, 2 members, incomplete)
multipolygon (1588158, 3 members) [id: 1,588,1
multipolygon (1588825, 2 members) [id: 1,588,8
multipolygon (1588864, 2 members) [id: 1,588,8
multipolygon (1588865, 3 members) [id: 1,588,8
multipolygon (1588866, 2 members) [id: 1,588,8
multipolygon (1594322, 2 members) [id: 1,594,3
multipolygon ("Gostiny Dvor", 4 members, incomp
multipolygon ("Hare Island", 2 members) [id: 1,57
multipolygon ("Васильевский остров", 20 membe
multipolygon ("Кронверкский пролив", 4 members,
multipolygon ("Малая Нева", 27 members, incomple
multipolygon ("Нева", 38 members, incomplete) [id:
multipolygon ("Петроградский остров", 14 membe
multipolygon ("Российский научный центр «Прикл
multipolygon[5] ("Василеостровский район", 27 me
multipolygon[5] ("Калининский район", 72 member
multipolygon[5] ("Красногвардейский район", 26
multipolygon[5] ("Петроградский район", 19 mem
multipolygon[5] ("Центральный район", 37 memb
multipolygon[8] ("Васильевский", 24 members, in
multipolygon[8] ("Введенский", 25 members, inco
multipolygon[8] ("Гавань", 24 members, incomple
multipolygon[8] ("Дворцовый", 6 members, incom
multipolygon[8] ("Кронверкское", 42 mem
multipolygon[8] ("Муниципальный круг № 7", 3
multipolygon[8] ("Пет ий округ", 54 member
multipolygon[№ 78", 16 members, incomplete
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restriction ("no_left_turn", 3 members) [id: 1,662,
restriction ("no_left_turn", 3 members, incomplete
restriction ("no_left_turn", 3 members, incomplete
restriction ("no_left_turn", 3 members, incomplete
restriction ("no_right_turn", 3 members, incomplet
restriction ("no_right_turn", 3 members, incomplet
restriction ("no_u_turn", 1 member) [id: 1,662,61
restriction ("only_right_turn", 3 members) [id: 408
restriction ("only_right_turn", 3 members) [id: 1,1
restriction ("only_right_turn", 3 members, incomple
restriction ("only_straight_on", 3 members) [id: 40
restriction ("only_straight_on", 3 members) [id: 1,
restriction ("only_straight_on", 3 members) [id: 1,1
restriction ("only_straight_on", 3 members) [id: 1,1
restriction ("only_straight_on", 3 members) [id: 1,6
restriction ("only_straight_on", 3 members) [id: 1,6
restriction ("only_straight_on", 3 members) [id: 1,6
restriction ("only_straight_on", 3 members) [id: 1,6
restriction ("only_straight_on", 3 members) [id: 1,6
restriction ("only_straight_on", 3 members) [id: 1,6
restriction ("only_straight_on", 3 members) [id: 1,6
restriction ("only_straight_on", 3 members, incomple
restriction ("only_straight_on", 3 members, incomple
restriction ("only_straight_on", 3 members, incomple
restriction ("only_straight_on", 3 members, incomple
restriction ("only_straight_on", 3 members, incomple
restriction ("only_straight_on", 3 members, incomple
route ("1", 12 members) [id: 1,625,924]
route ("1", 12 members) [id: 1,625,925]
route ("1", 109 members, incomplete) [id: 946,411]
route ("1", 114 members, incomplete) [id: 969,924]
route ("1", 116 members, incomplete) [id: 960,841]
route ("1", 127 members, incomplete) [id: 969,270]
route ("3", 78 members, incomplete) [id: 969,921]
route ("3", 84 members, incomplete) [id: 969,636]
route ("5", 51 members, incomplete) [id: 969,640]
route ("5", 58 members, incomplete) [id: 969,631]
route ("6", 34 members, incomplete) [id: 569,976]
route ("6", 41 members, incomplete) [id: 1,624,636
route ("6", 90 members, incomplete) [id: 962,194]

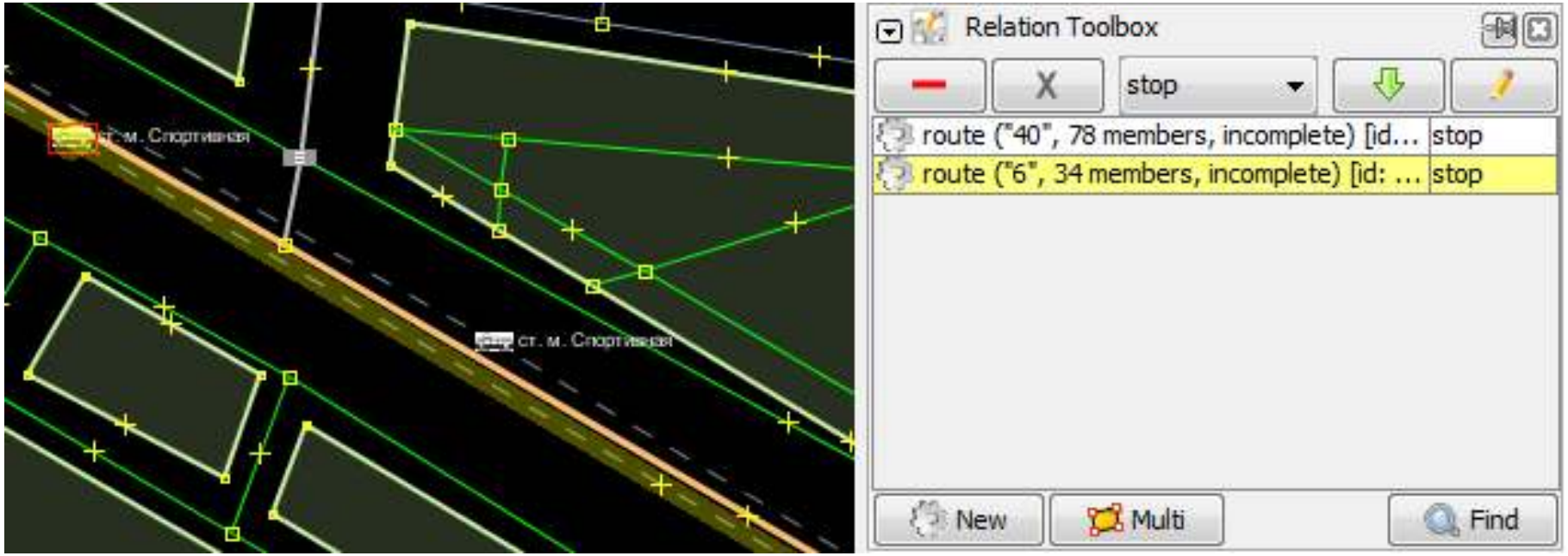
NO WAI

Members → Relations



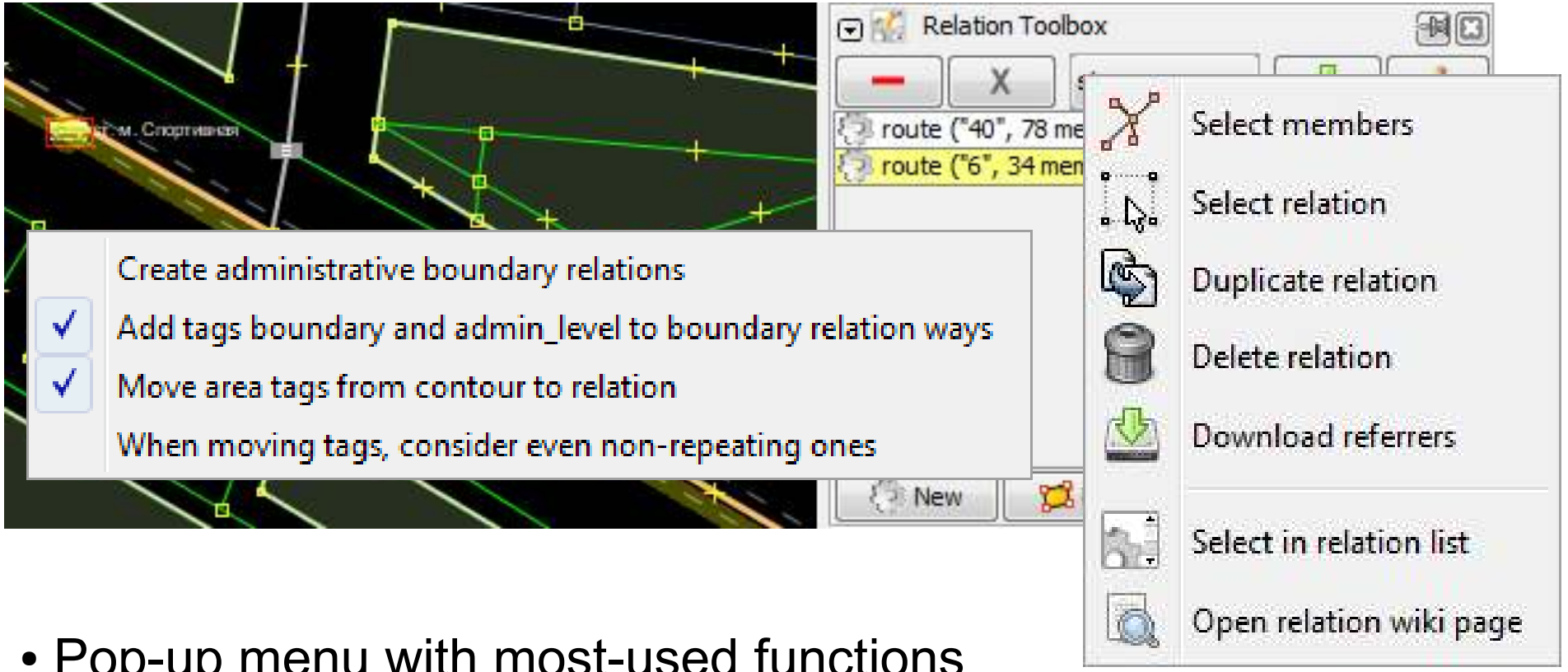


Relation Toolbox



- *Chosen relation*: independent from selection
- It is highlighted, members are visible without lists
- Simple add/remove button and role chooser
- Service buttons: edit, create new, download incomplete etc.

Relation Toolbox



- Pop-up menu with most-used functions
- Multipolygon properties at hand
- When integrated, the plugin can be reduced to buttons in toolbar

What can be improved

- Status of chosen relation: number of rings and segments, are members sorted
- «Sort» button and a dialog for arranging members
- Relation-specific contextual functions: find stops for PT route; wizards for creating relations
- Better visualization on the map: color-code area relations, highlight errors

So, to the point

- Relations are scary, because editing them is uncomfortable
- This can be fixed
- Users don't care for data structures, they need a button
- Make user-friendly relation tools
- Example: Relation Toolbox

Thanks!